AFRICAST

TIDY TIME SERIES & FORECASTING IN R

Forecasting with regression

Representing temporal structure with regressors

Session 5 of 10



SECTION 1 OF 6

The linear model with time series

1 **The linear model with time series**
2 Evaluating the regression model
3 Selecting predictors
4 Forecasting with regression
5 Correlation, causation and forecasting
6 Some useful predictors for regression models



Regression models

- To **explain**
- To **forecast**
- Simple linear regression model (SLR)
- Multiple linear regression model (MLR)

SLR model in theory

Regression model allows for a linear relationship between the forecast variable y and a single predictor variable x.

$$y_t = \beta_0 + \beta_1 x_t + \varepsilon_t.$$

- y_t is the variable we want to predict: the “response” variable
- Each x_t is numerical and is called a “predictor”
- β_0 and β_1 are regression coefficients

SLR model in practice

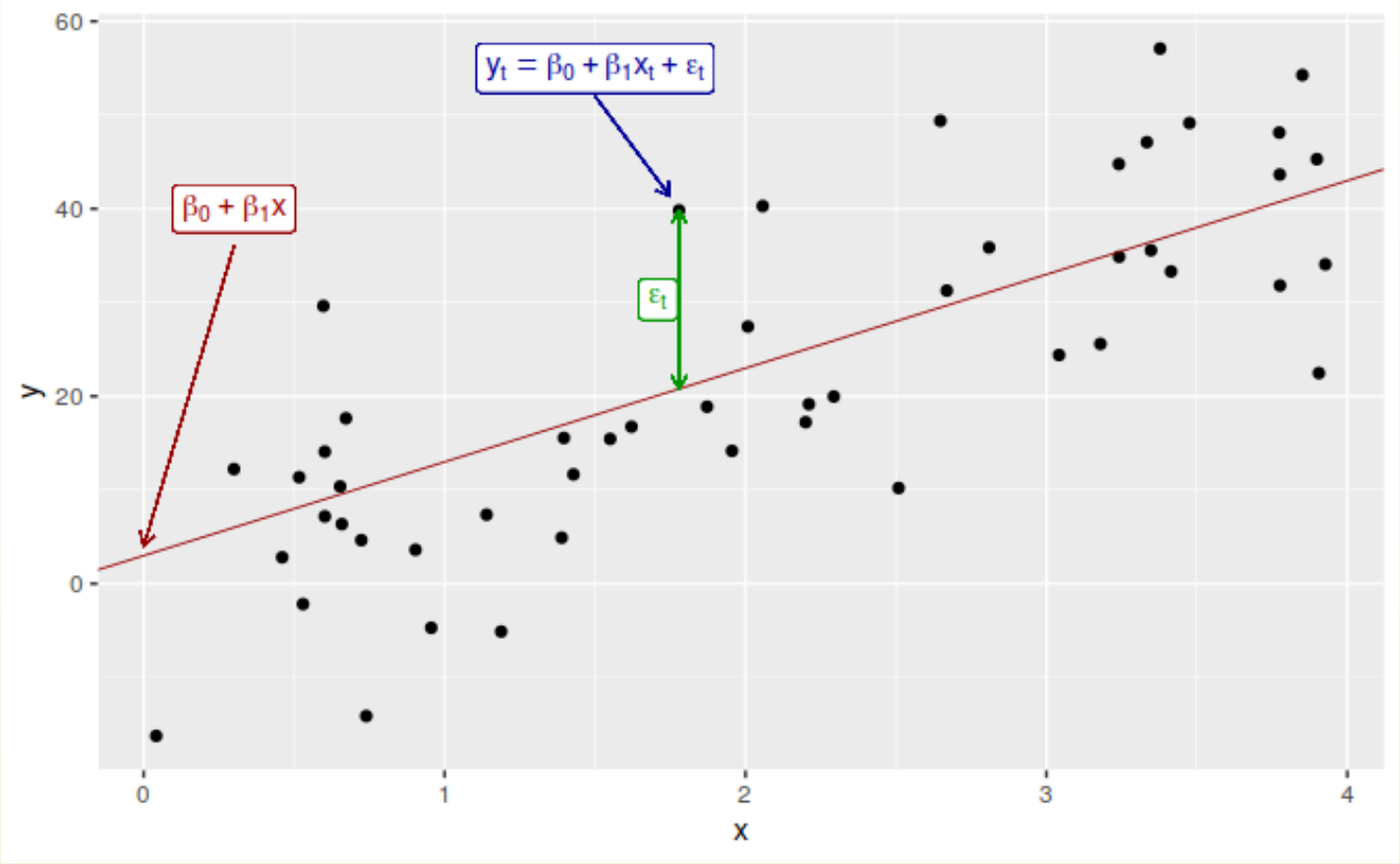
In practice, of course, we have a collection of observations but we do not know the values of the coefficients β_0, β_1. These need to be estimated from the data.

$$\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 x_t.$$

- $\hat{y}_t$ is the fitted value of the response
- Each x_t is a "predictor"
- $\hat{\beta}_0$ is the estimated intercept
- $\hat{\beta}_1$ is the estimated slope

What is the best fit

- There are many ways that a straight line can be laid on the scatter
- Best known criterion is called Ordinary Least Squares (OLS)



Estimation of the model

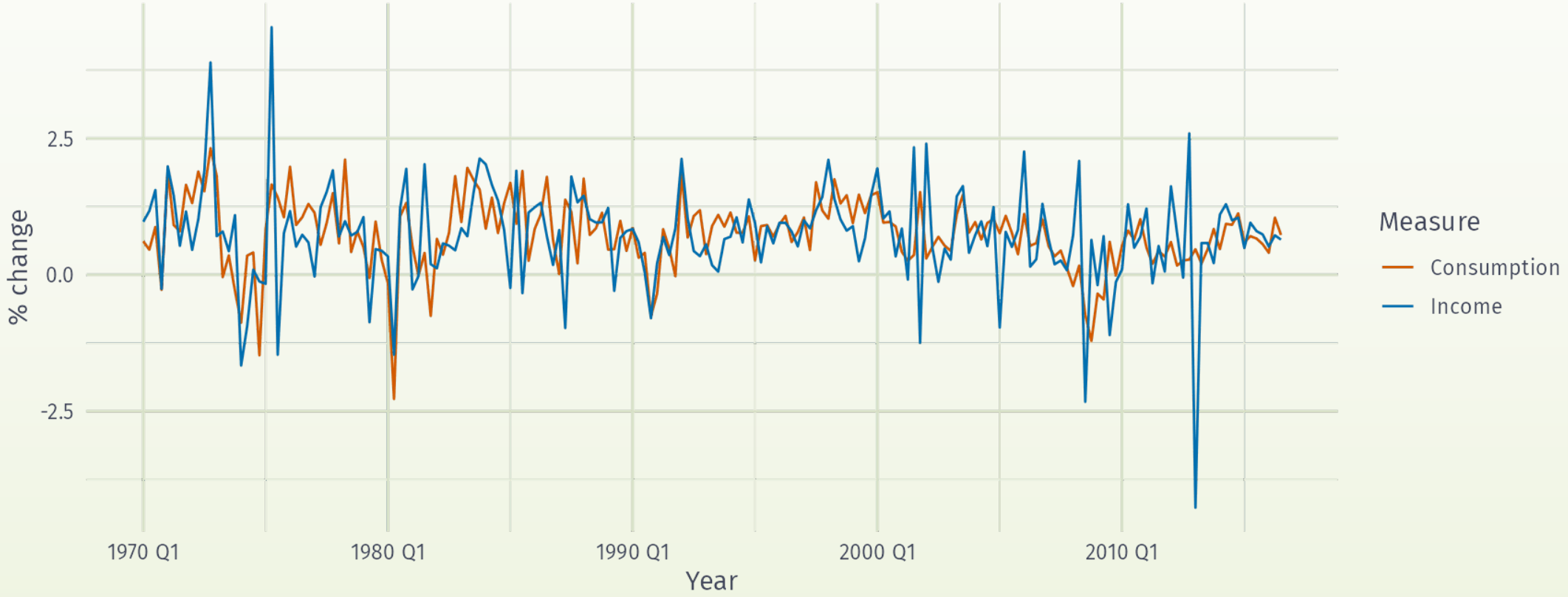
Least squares chooses the values of β_0 and β_1 which minimise

$$\sum_{t=1}^{T} e_t^2 = \sum_{t=1}^{T} (y_t - \beta_0 - \beta_1 x_t)^2.$$

- This is called *least squares* estimation because it gives the smallest value of the sum of squared errors.
- Finding the best estimates of the coefficients is often called *fitting* the model to the data.
- We refer to the *estimated* coefficients using the notation $\hat{\beta}_0, \hat{\beta}_1$.

Example: US consumption expenditure

```
us_change |>
  pivot_longer(c(Consumption, Income), names_to = "Measure", values_to = "Change") |>
  autoplot(Change) +
  labs(x = "Year", y = "% change")
```



Example: US consumption expenditure



Example: US consumption expenditure

```
fit_income <- us_change |>
  model(lm = TSLM(Consumption ~ Income))
report(fit_income)
```

```
Series: Consumption
Model: TSLM

Residuals:
   Min     1Q Median     3Q    Max
-2.408 -0.318  0.026  0.300  1.452

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.5451     0.0557    9.79  < 2e-16 ***
Income        0.2806     0.0474    5.91  1.6e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.603 on 185 degrees of freedom
Multiple R-squared: 0.159,  Adjusted R-squared: 0.154
F-statistic:   35 on 1 and 185 DF, p-value: 2e-08
```

Interpretation

A 1 percentage point rise in income growth goes with a 0.28 point rise in consumption growth.

Income alone explains little: $R^2 = 0.16$.

Multiple regression

- In multiple regression there is one variable to be forecast and several predictor variables.
- The basic concept is that we forecast the time series of interest y assuming that it has a linear relationship with other time series x_1, x_2, ..., x_k
- We might forecast daily A&E attendance y using temperature x_1 and GP visits x_2 as predictors.

How many predictors?

You can add as many as you want but be aware of:

- Over-fitting
- Multicollinearity

Multiple regression and forecasting

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \cdots + \beta_k x_{k,t} + \varepsilon_t.$$

- y_t is the variable we want to predict: the "response" variable
- Each $x_{j,t}$ is numerical and is called a "predictor". They are usually assumed to be known for all past and future times.
- The coefficients $\beta_1, \ldots, \beta_k$ measure the effect of each predictor after taking account of the effect of all other predictors in the model.
  - That is, the coefficients measure the **marginal effects**.
- ε_t is a white noise error term

Estimation of the model

Least squares chooses the values of $\beta_0, \ldots, \beta_k$ which minimise

$$\sum_{t=1}^{T} e_t^2 = \sum_{t=1}^{T} (y_t - \beta_0 - \beta_1 x_{1,t} - \cdots - \beta_k x_{k,t})^2.$$

- This is called *least squares* estimation because it gives the smallest value of the sum of squared errors
- Finding the best estimates of the coefficients is often called *fitting* the model to the data
- We refer to the *estimated* coefficients using the notation $\hat{\beta}_0, \ldots, \hat{\beta}_k$.

Useful predictors in linear regression

Linear trend

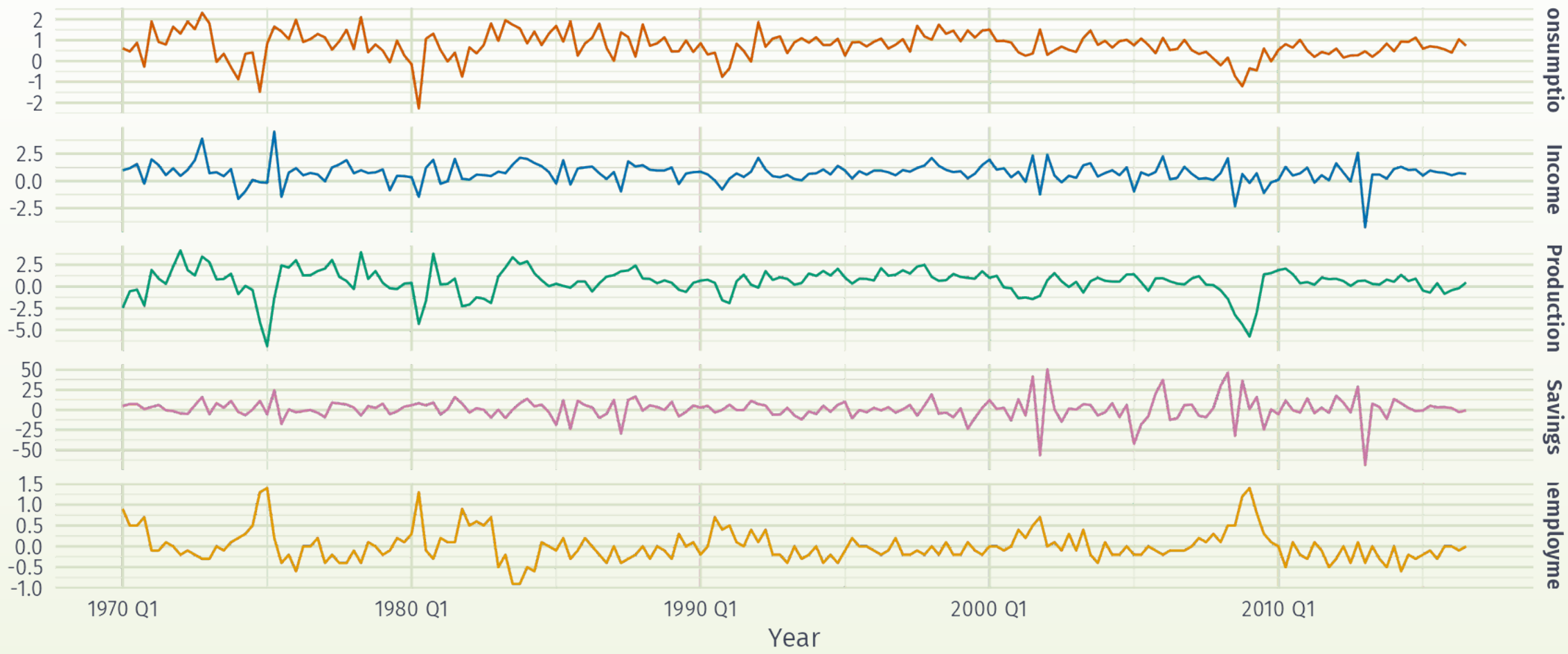
$$x_t = t$$

- $t = 1, 2, \ldots, T$
- Strong assumption that trend will continue.
- use special function `trend()`

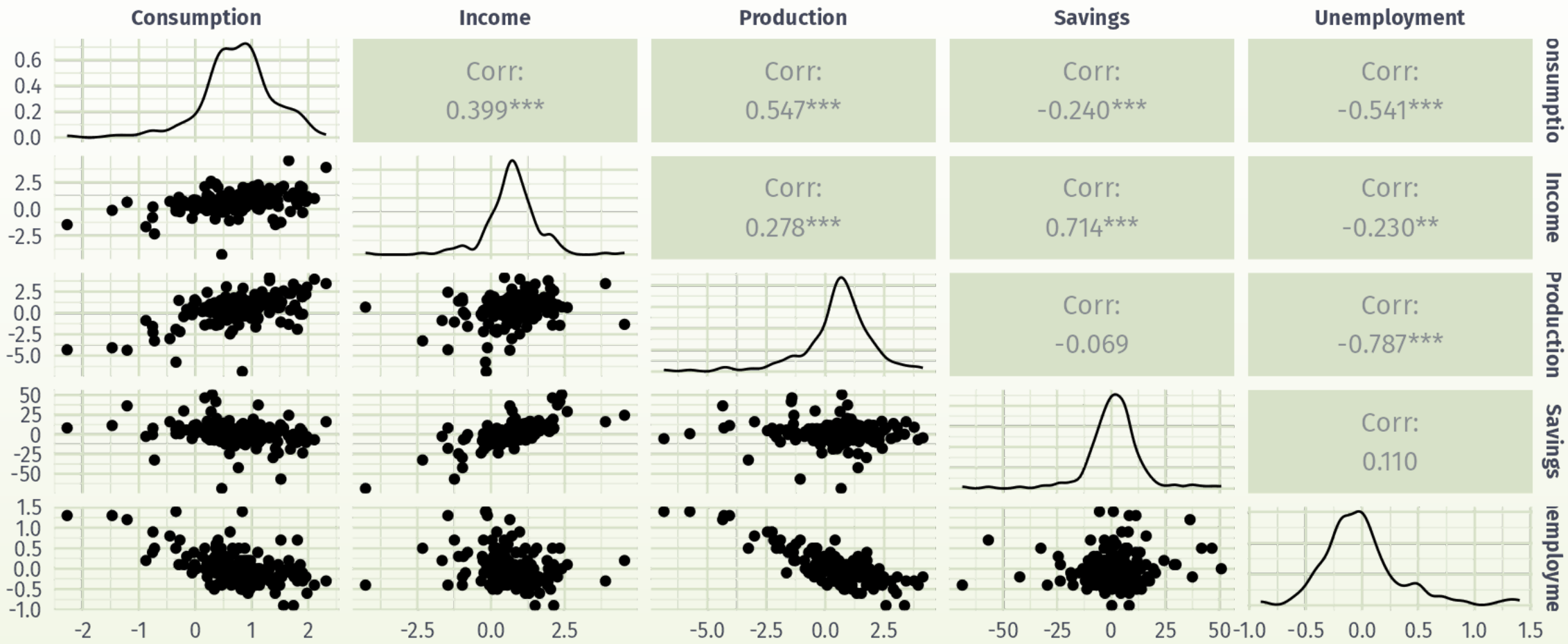
Seasonality

- Seasonality will be considered based on the interval of index
- use special function `season()`

Example: US consumption expenditure



Example: US consumption expenditure



Example: US consumption expenditure

```
fit_full <- us_change |>
  model(lm = TSLM(Consumption ~ Income + Production + Unemployment + Savings))
report(fit_full)
```

```
Series: Consumption
Model: TSLM

Residuals:
   Min     1Q Median     3Q    Max
-0.883 -0.176 -0.037  0.153  1.206

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.26729    0.03721    7.18  1.7e-11 ***
Income        0.71448    0.04219   16.93  < 2e-16 ***
Production    0.04589    0.02588    1.77    0.078 .
Unemployment -0.20477    0.10550   -1.94    0.054 .
Savings      -0.04527    0.00278  -16.29  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.329 on 182 degrees of freedom
Multiple R-squared: 0.754,  Adjusted R-squared: 0.749
F-statistic:  139 on 4 and 182 DF, p-value: <2e-16
```

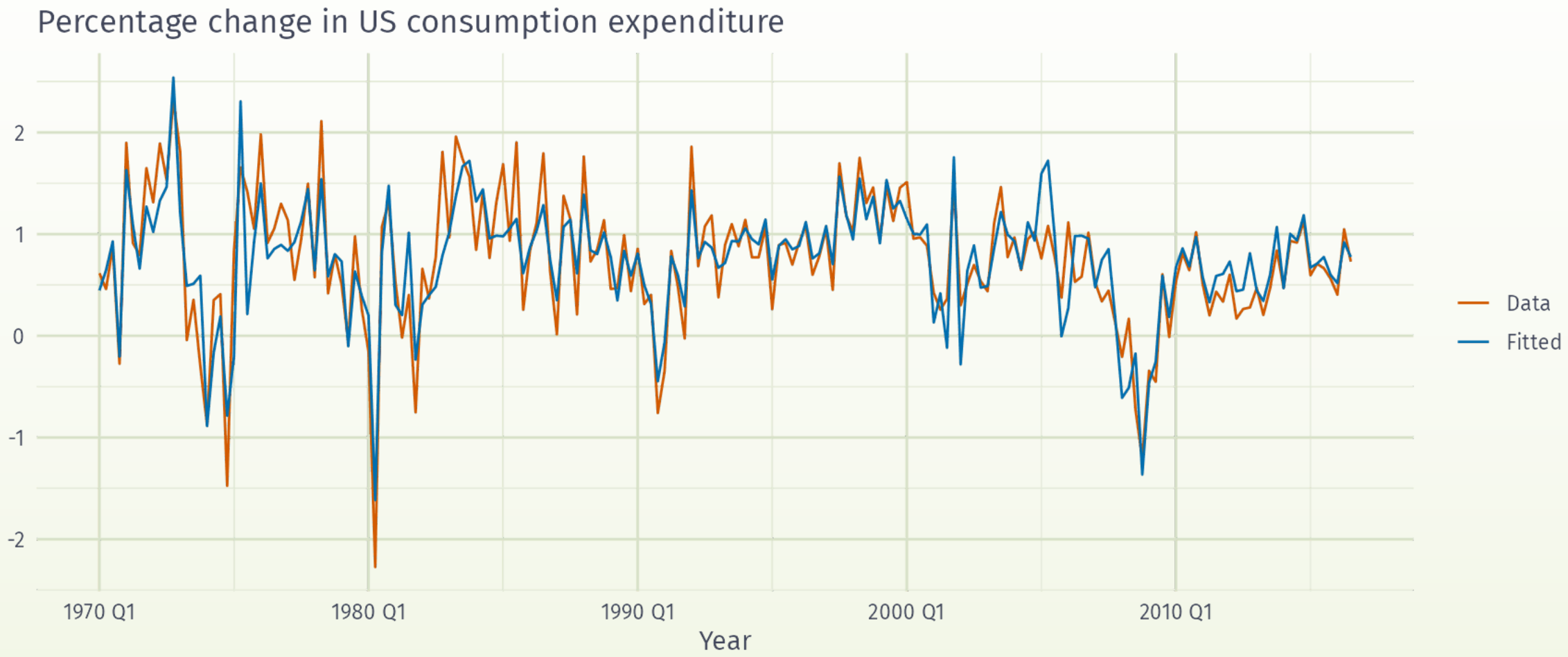
Interpretation

Each coefficient holds the other predictors fixed.

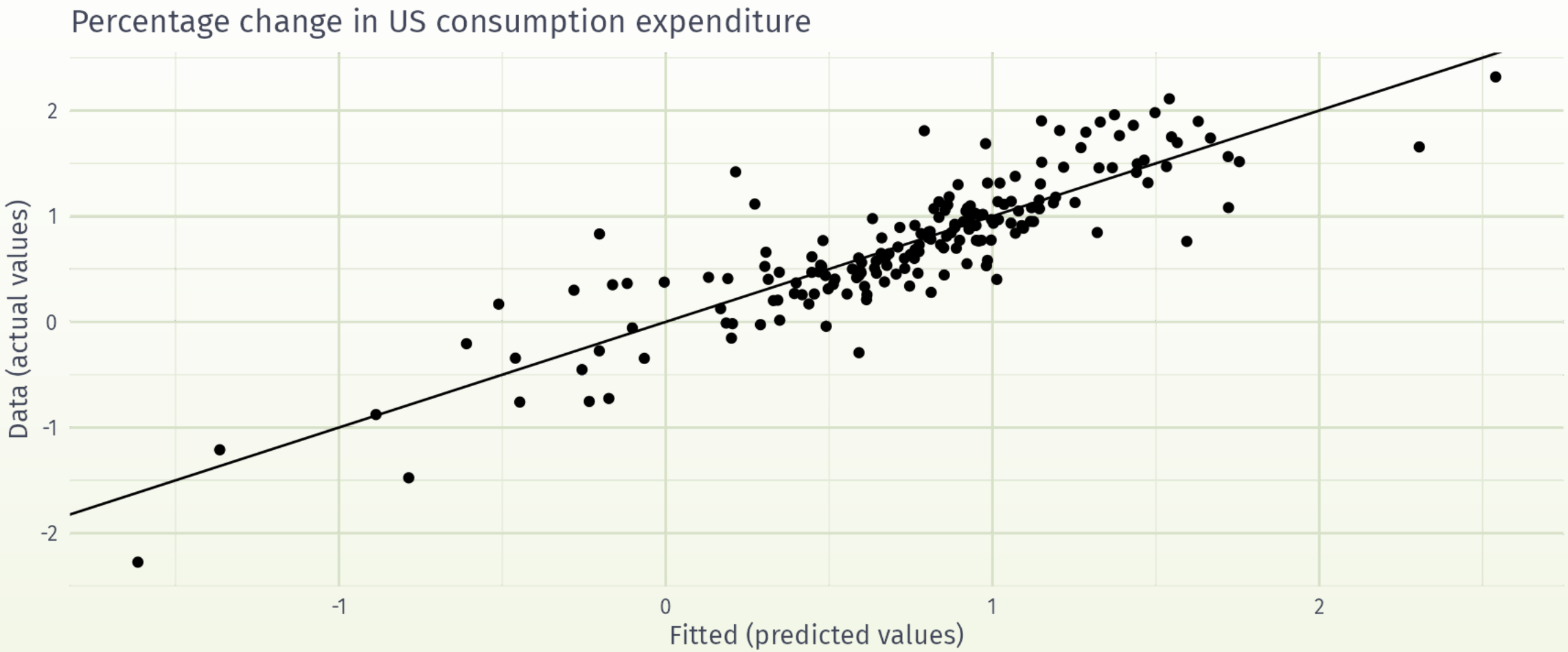
Production and unemployment are borderline ($p \approx 0.05$ to 0.08).

$\bar{R}^2 = 0.75$, far above income alone.

Example: US consumption expenditure



Example: US consumption expenditure



SECTION 2 OF 6

Evaluating the regression model

1 The linear model with time series
2 **Evaluating the regression model**
3 Selecting predictors
4 Forecasting with regression
5 Correlation, causation and forecasting
6 Some useful predictors for regression models



Assumptions for forecasting

For forecasting purposes, we require the following assumptions:

- ε_t are uncorrelated and zero mean
- ε_t are uncorrelated with each $x_{j,t}$.

It is **useful** to also have $\varepsilon_t \sim \mathrm{N}(0, \sigma^2)$ when producing prediction intervals or doing statistical tests.

Residual diagnostics

There are a series of plots that should be produced in order to check different aspects of the fitted model and the underlying assumptions.

1. check if residuals are uncorrelated using `ACF`
2. Check if residuals are normally distributed

Ljung-Box test

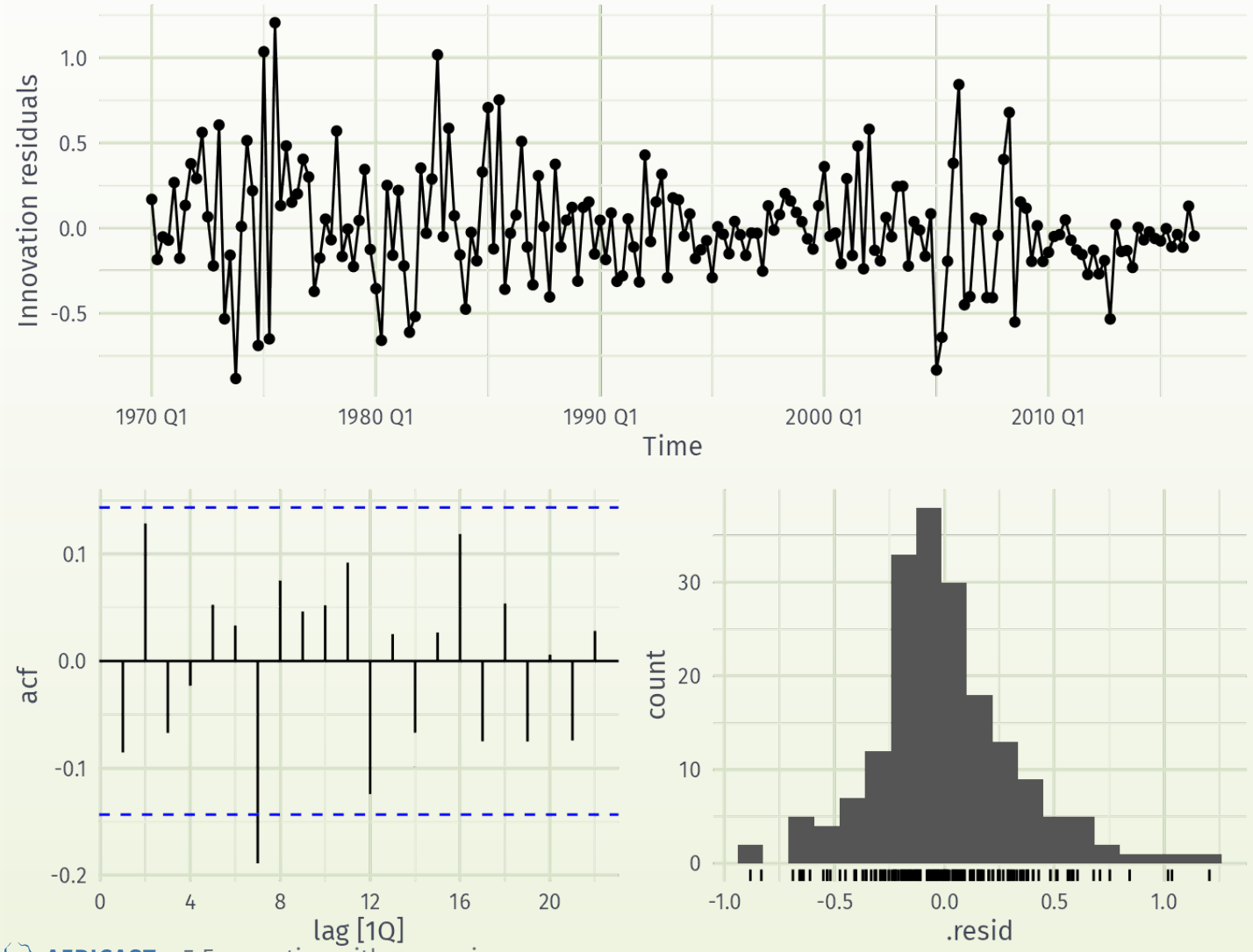
Test whether *whole set* of r_k values is significantly different from zero set.

$$Q = T(T+2)\sum_{k=1}^{\ell}(T-k)^{-1}r_k^2 \qquad \text{where } \ell = \text{max lag and } T = \#\text{observations}$$

- If each r_k close to zero, Q will be **small**.
- If some r_k values large ($+$ or $-$), Q will be **large**.
- Suggested lags: $\ell = 10$ for non-seasonal data, $\ell = 2m$ for seasonal data (but at most $T/5$).
- If data are WN and T large, $Q \sim \chi^2$ with ℓ degrees of freedom.
- For residuals of a model with K estimated parameters, use $\ell - K$ degrees of freedom (`dof` argument).

Example: US consumption expenditure

```
gg_tsresiduals(fit_full)
```



Interpretation

A few ACF spikes outside the bounds (Ljung-Box $p = 0.073$), and a slightly skewed histogram.

Not perfect, but fine for forecasting. Intervals may be slightly off.

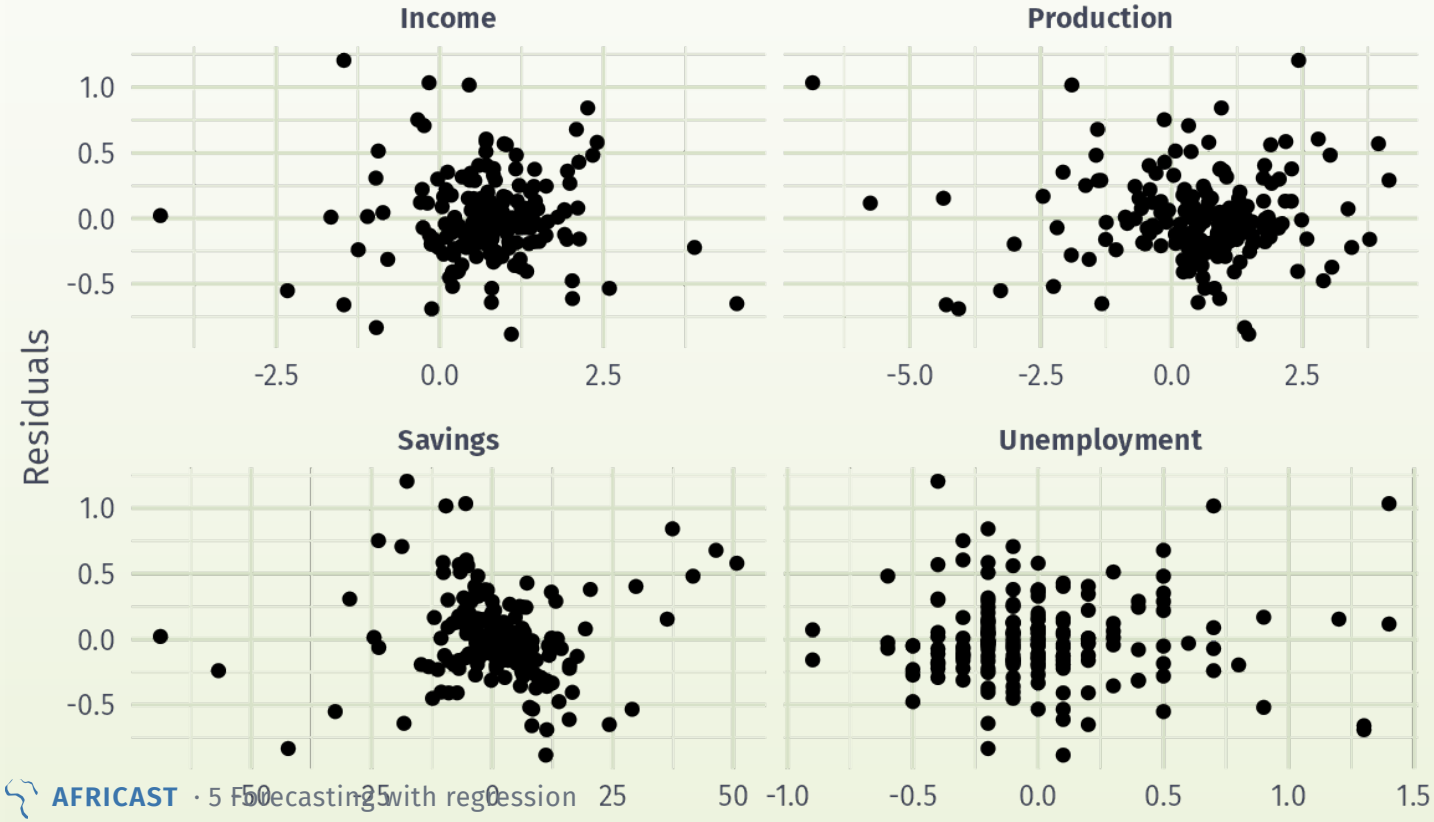
Residual scatterplots

Useful for spotting outliers and whether the linear model was appropriate.

- Scatterplot of residuals e_t against each predictor $x_{j,t}$.
- Scatterplot residuals against the fitted values $\hat{y}_t$
- Expect to see scatterplots resembling a horizontal band with no values too far from the band and no patterns such as curvature or increasing spread.

Example: US consumption expenditure

```
us_change |>
  left_join(residuals(fit_full), by = "Time") |>
  pivot_longer(c(Income, Production, Savings, Unemployment)) |>
  ggplot(aes(x = value, y = .resid)) +
  geom_point() +
  facet_wrap(vars(name), scales = "free_x") +
  labs(x = NULL, y = "Residuals")
```



Interpretation

No curvature or fanning: linear terms look adequate.

Residual patterns

- If a plot of the residuals vs any predictor in the model shows a pattern, then the relationship is non-linear.
- If a plot of the residuals vs any predictor **not** in the model shows a pattern, then the predictor should be added to the model.
- If a plot of the residuals vs fitted values shows a pattern, then there is heteroscedasticity in the errors. (Could try a transformation.)

SECTION 3 OF 6

Selecting predictors

1 The linear model with time series
2 Evaluating the regression model
3 **Selecting predictors**
4 Forecasting with regression
5 Correlation, causation and forecasting
6 Some useful predictors for regression models



Comparing regression models

Computer output for regression will always give the R^2 value. This is a useful summary of the model.

- It is equal to the square of the correlation between y and $\hat{y}$.
- It is often called the "coefficient of determination".
- It can also be calculated as follows: $R^2 = \frac{\sum(\hat{y}_t - \bar{y})^2}{\sum(y_t - \bar{y})^2}$
- It is the proportion of variance accounted for (explained) by the predictors.

Adding predictors

Adding a useless predictor to a regression model makes R^2...

Increase or stay the same	0	0%
Decrease	0	0%
Change unpredictably	0	0%



poll.nectric.com.au/s/
africast-regression

0 responses

● CLOSED 0 connected

Comparing regression models

However ...

- R^2 does not penalise model size.
- Adding *any* variable tends to increase the value of R^2, even if that variable is irrelevant.

To overcome this problem, we can use *adjusted* R^2:

$$\bar{R}^2 = 1 - (1 - R^2)\frac{T-1}{T-k-1}$$

where $k =$ no. predictors and $T =$ no. observations.

Maximising $\bar{R}^2$ is equivalent to minimising $\hat{\sigma}^2$.

$$\hat{\sigma}^2 = \frac{1}{T-k-1}\sum_{t=1}^{T} e_t^2$$

Cross-validation

1. Remove observation t from the data set, and fit the model using the remaining data. Then compute the error for the omitted observation
2. Repeat step 1 for $t = 1, \ldots, T$
3. Compute the MSE from errors obtained in 1. We shall call this the CV

Akaike's Information Criterion

$$\text{AIC} = -2\log(L) + 2(k+2)$$

where L is the likelihood and k is the number of predictors in the model.

- This is a *penalised likelihood* approach.
- *Minimising* the AIC tends to give the best model for prediction.
- $\bar{R}^2$ tends to select too many predictors, so prefer CV, AIC or AICc.
- Minimising the AIC is asymptotically equivalent to minimising MSE via leave-one-out cross-validation.

Corrected AIC

For small values of T, the AIC tends to select too many predictors, and so a bias-corrected version of the AIC has been developed.

$$\text{AIC}_\text{C} = \text{AIC} + \frac{2(k+2)(k+3)}{T-k-3}$$

As with the AIC, the AIC_C should be minimised.

Choosing regression variables

Best subsets regression

- Fit all possible regression models using one or more of the predictors.
- Choose the best model based on one of the measures of predictive ability (CV, AIC, AICc).

Choosing regression variables

Backwards stepwise regression

- Start with a model containing all variables.
- Try subtracting one variable at a time. Keep the model if it has lower CV or AICc.
- Iterate until no further improvement.
- You can also do forward stepwise

Example: candidate models

Start from the full model, then drop one predictor at a time.

```
fit_candidates <- us_change |>
  model(
    full = TSLM(Consumption ~ Income + Production + Unemployment + Savings),
    no_income = TSLM(Consumption ~ Production + Unemployment + Savings),
    no_production = TSLM(Consumption ~ Income + Unemployment + Savings),
    no_unemployment = TSLM(Consumption ~ Income + Production + Savings),
    no_savings = TSLM(Consumption ~ Income + Production + Unemployment)
  )
```

Example: candidate models

Which predictor can we most afford to drop from the full model?

Income	0	0%
Production	0	0%
Unemployment	0	0%
Savings	0	0%



poll.nectric.com.au/s/africast-regression

0 responses

● CLOSED 0 connected

Example: comparing candidate models

```
glance(fit_candidates) |>
  select(.model, adj_r_squared, CV, AIC, AICc) |>
  arrange(AICc)
```

```
# A tibble: 5 × 5
  .model          adj_r_squared    CV   AIC  AICc
  <chr>                   <dbl> <dbl> <dbl> <dbl>
1 full                    0.749 0.116 -409. -409.
2 no_production           0.746 0.116 -408. -408.
3 no_unemployment         0.745 0.118 -407. -407.
4 no_savings              0.386 0.278 -243. -243.
5 no_income               0.356 0.293 -234. -234.
```

Interpretation

The full model has the lowest AICc and CV, so keep all four predictors.

Dropping production is almost as good; dropping income or savings is much worse.

SECTION 4 OF 6

Forecasting with regression

1 The linear model with time series
2 Evaluating the regression model
3 Selecting predictors
4 **Forecasting with regression**
5 Correlation, causation and forecasting
6 Some useful predictors for regression models



Ex-ante versus ex-post forecasts

- *Ex ante forecasts* are made using only information available in advance.
  - require forecasts of predictors
- *Ex post forecasts* are made using later information on the predictors.
  - useful for studying behaviour of forecasting models.
- trend, seasonal and calendar variables are all known in advance, so these don't need to be forecast.

Scenario based forecasting

- Assumes possible scenarios for the predictor variables
- Prediction intervals for scenario based forecasts do not include the uncertainty associated with the future values of the predictor variables.

Building a predictive regression model

- If getting forecasts of predictors is difficult, you can use lagged predictors instead.

$$y_t = \beta_0 + \beta_1 x_{1,t-h} + \cdots + \beta_k x_{k,t-h} + \varepsilon_t.$$

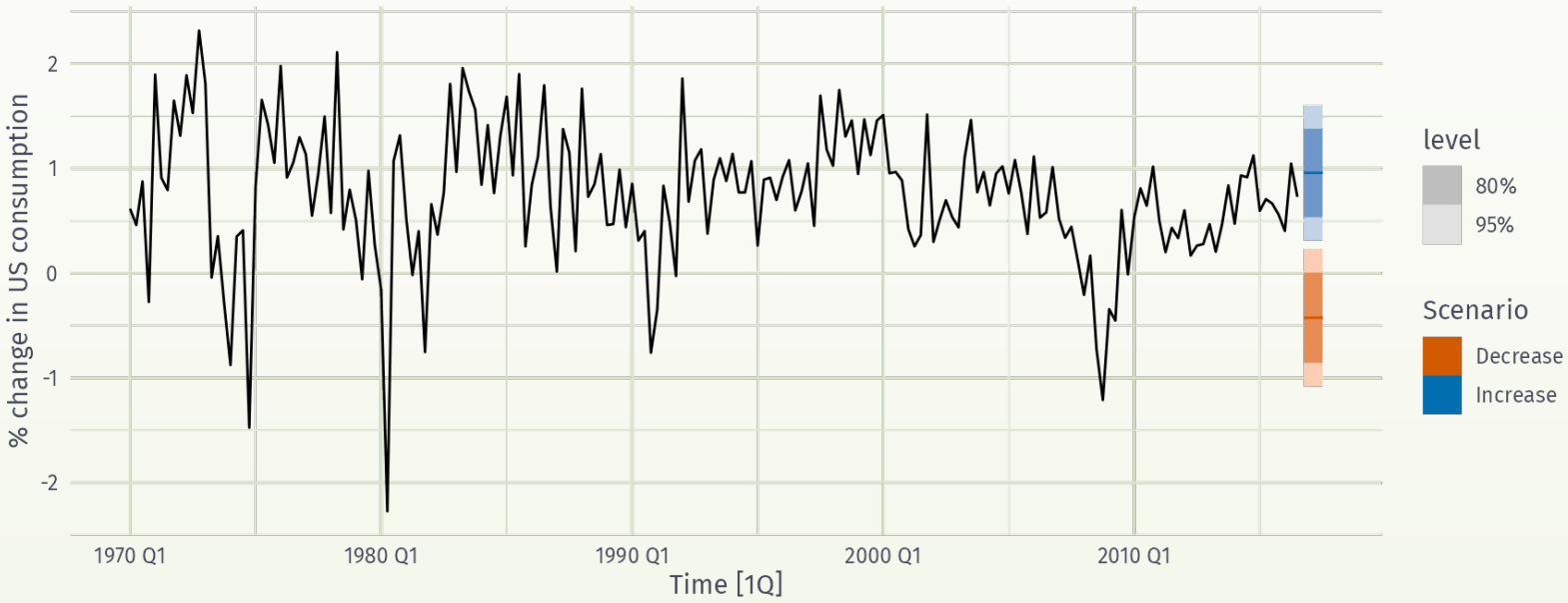
- A different model for each forecast horizon h.

US Consumption

```
future_scenarios <- scenarios(
  Increase = new_data(us_change, 4) |>
    mutate(Income = 1, Savings = 0.5, Unemployment = 0, Production = 0),
  Decrease = new_data(us_change, 4) |>
    mutate(Income = -1, Savings = -0.5, Unemployment = 0, Production = 0),
  names_to = "Scenario"
)
fc_scenarios <- forecast(fit_full, new_data = future_scenarios)
```

US Consumption

```
us_change |> autoplot(Consumption) +
  autolayer(fc_scenarios) +
  labs(y = "% change in US consumption")
```



Interpretation

Scenarios shift the point forecasts. The intervals ignore uncertainty in the future predictors.

SECTION 5 OF 6

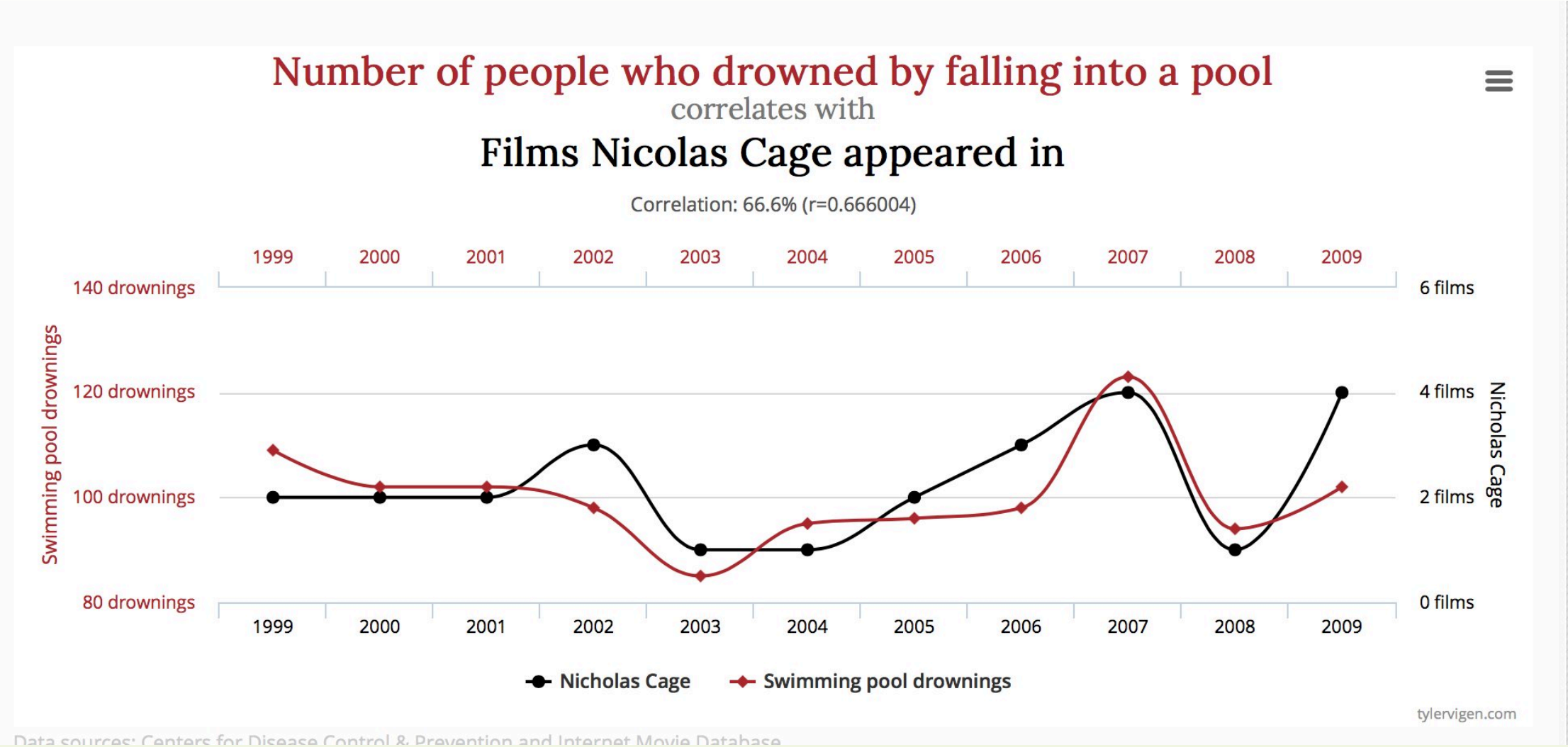
Correlation, causation and forecasting

1 The linear model with time series
2 Evaluating the regression model
3 Selecting predictors
4 Forecasting with regression
5 **Correlation, causation and forecasting**
6 Some useful predictors for regression models



Correlation does not imply causation

Check out https://www.tylervigen.com/spurious-correlations



Correlation and forecasting

Ice-cream sales are strongly correlated with drownings. Are ice-cream sales useful for forecasting drownings?



poll.nectric.com.au/s/africast-regression

0 responses

CLOSED 0 connected

Correlation is not causation

- When x is useful for predicting y, it is not necessarily causing y.
- e.g., predict number of drownings y using number of ice-creams sold x.
- Correlations are useful for forecasting, even when there is no causality.
- Better models usually involve causal relationships (e.g., temperature and number of swimmers to predict drownings).

Multicollinearity

In regression analysis, multicollinearity occurs when:

- Two predictors are highly correlated (i.e., the correlation between them is close to ± 1).
- A linear combination of some of the predictors is highly correlated with another predictor.
- A linear combination of one subset of predictors is highly correlated with a linear combination of another subset of predictors.

Multicollinearity

If multicollinearity exists...

- coefficient estimates become unstable and have large standard errors
- don't rely on the p-values to determine significance.
- there is no problem with model *predictions* provided the predictors used for forecasting are within the range used for fitting.
- omitting variables can help.
- combining variables can help.

Outliers and influential observations

Things to watch for

- *Outliers*: observations that produce large residuals.
- *Influential observations*: removing them would markedly change the coefficients. (Often outliers in the x variable).
- *Lurking variable*: a predictor not included in the regression but which has an important effect on the response.
- Points should not normally be removed without a good explanation of why they are different.

Modern regression models

- Suppose we had 44 predictors instead of 4.
  - That gives $2^{44} \approx 18$ trillion possible models!
- Best subsets and stepwise search become infeasible.
- Shrinkage methods (lasso, ridge, elastic net) handle this, but are beyond this workshop.

SECTION 6 OF 6

Some useful predictors for regression models

1 The linear model with time series
2 Evaluating the regression model
3 Selecting predictors
4 Forecasting with regression
5 Correlation, causation and forecasting
6 **Some useful predictors for regression models**



Dummy variables

If a categorical variable takes only two values (e.g., “Yes” or “No”), then an equivalent numerical variable can be constructed taking value 1 if yes and 0 if no. This is called a **dummy variable**.

Market day?
Yes
Yes
No
Yes
No
No
Yes
No

recode →

d_t
1
1
0
1
0
0
1
0

Yes → 1, No → 0

Dummy variables

If there are more than two categories, then the variable can be coded using several dummy variables (one fewer than the total number of categories).

Day	$d_{\text{Mon},t}$	$d_{\text{Tue},t}$	$d_{\text{Wed},t}$	$d_{\text{Thu},t}$
Monday	1	0	0	0
Tuesday	0	1	0	0
Wednesday	0	0	1	0
Thursday	0	0	0	1
Friday	0	0	0	0
Monday	1	0	0	0
Tuesday	0	1	0	0
Wednesday	0	0	1	0
Thursday	0	0	0	1
Friday	0	0	0	0

← **Friday: all zeros (the baseline)**

5 categories → 4 dummies

Beware of the dummy variable trap!

- Using one dummy for each category gives too many dummy variables!
- The regression will then be singular and inestimable.
- Either omit the constant, or omit the dummy for one category.
- The coefficients of the dummies are relative to the omitted category.

Uses of dummy variables

Seasonal dummies

- For quarterly data: use 3 dummies
- For monthly data: use 11 dummies
- For daily data: use 6 dummies

Outliers

- If there is an outlier, you can use a dummy variable to remove its effect.

Public holidays

- For daily data: if it is a public holiday, dummy=1, otherwise dummy=0.

Intervention variables

Spikes

- Equivalent to a dummy variable for handling an outlier.

Steps

- Variable takes value 0 before the intervention and 1 afterwards.

Change of slope

- Variables take values 0 before the intervention and values $\{1, 2, 3, \ldots\}$ afterwards.
- this could also be handled using a piecewise trend, `trend(knots = ...)`

Special events as dummies

- Christmas Eve: if Christmas Eve, $v_t = 1$, $v_t = 0$ otherwise
- New year's Day: if New year's Day, $v_t = 1$, $v_t = 0$ otherwise.
- and more: Ramadan and Chinese new year, school holiday, etc

Interactions

For example, sometimes the effect of a particular event might be different if it is on a weekend or a week day or its effect might be different in each shift:

- you need to introduce an interaction variable
- in a model formula, `v1 * v2` includes both main effects and their interaction `v1:v2`

Lagged predictors

The model includes present and past values of the predictor: $x_t, x_{t-1}, x_{t-2}, \ldots$.

$$y_t = \beta_0 + \gamma_0 x_t + \gamma_1 x_{t-1} + \cdots + \gamma_k x_{t-k} + \varepsilon_t$$

- x can influence y, but y is not allowed to influence x.

Lagged predictors

- Lagged values of a predictor:
  - Create new variables by shifting the existing variable backwards

Example: x is advertising which has a delayed effect

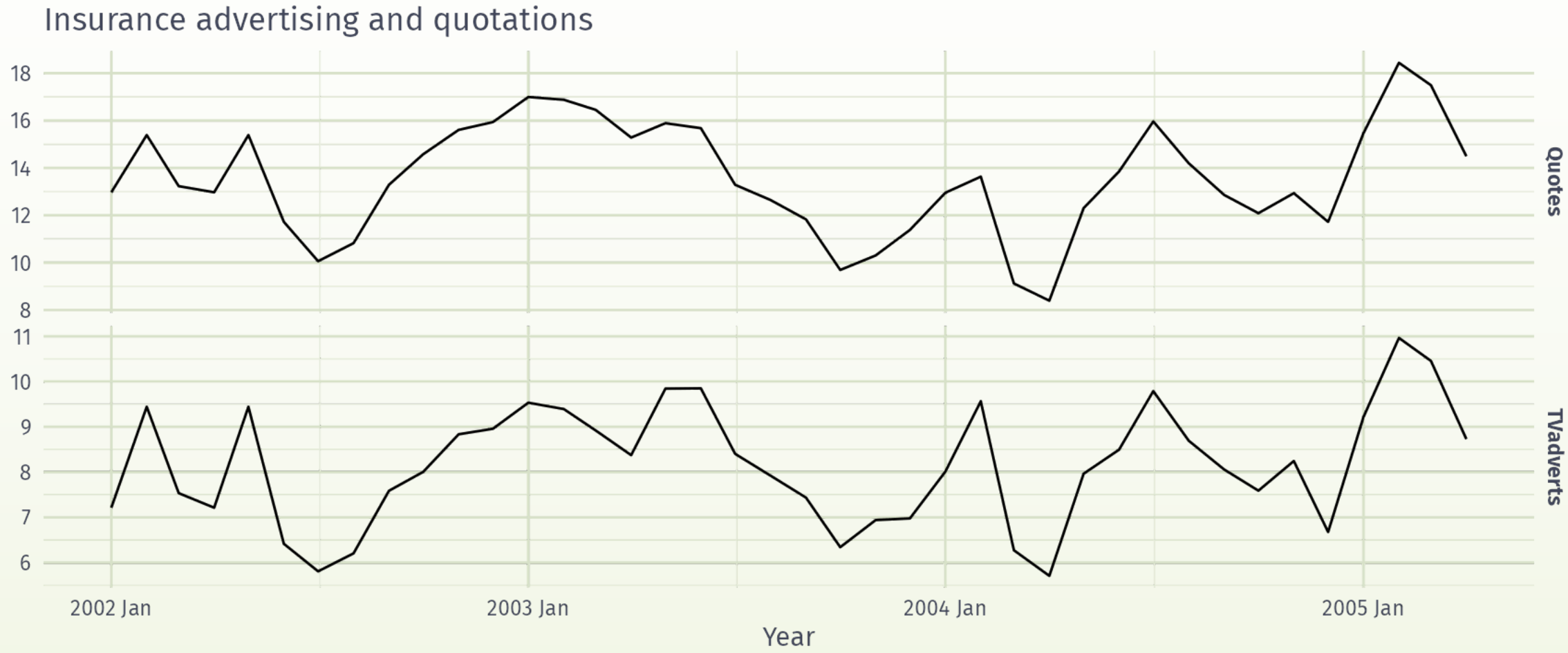
$$
\begin{aligned}
x_1 &= \text{advertising for previous month;} \\
x_2 &= \text{advertising for two months previously;} \\
&\vdots \\
x_m &= \text{advertising for } m \text{ months previously.}
\end{aligned}
$$

Example: Insurance quotes and TV adverts

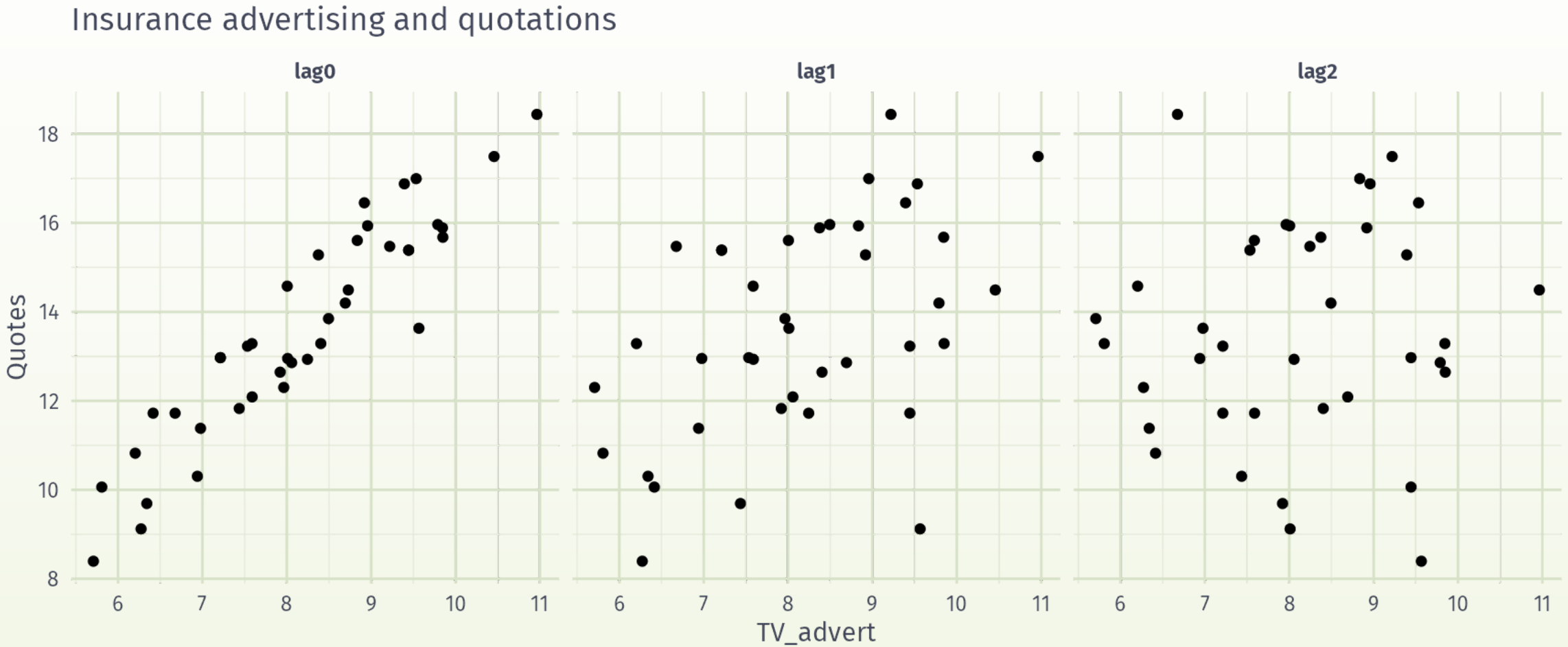
```
insurance
```

```
# A tsibble: 40 x 3 [1M]
      Month Quotes TVadverts
      <mth>  <dbl>     <dbl>
 1 2002 Jan   13.0      7.21
 2 2002 Feb   15.4      9.44
 3 2002 Mar   13.2      7.53
 4 2002 Apr   13.0      7.21
 5 2002 May   15.4      9.44
 6 2002 Jun   11.7      6.42
 7 2002 Jul   10.1      5.81
 8 2002 Aug   10.8      6.20
 9 2002 Sep   13.3      7.59
10 2002 Oct   14.6      8.00
# i 30 more rows
```

Example: Insurance quotes and TV adverts



Example: Insurance quotes and TV adverts



Example: Insurance quotes and TV adverts

```
fit_lags <- insurance |>
  # Restrict data so models use same fitting period
  mutate(Quotes = c(NA, NA, NA, Quotes[4:40])) |>
  model(
    lag0 = TSLM(Quotes ~ TVadverts),
    lag1 = TSLM(Quotes ~ TVadverts + lag(TVadverts)),
    lag2 = TSLM(Quotes ~ TVadverts + lag(TVadverts) + lag(TVadverts, 2)),
    lag3 = TSLM(Quotes ~ TVadverts + lag(TVadverts) + lag(TVadverts, 2) +
      lag(TVadverts, 3))
  )
```

Example: Insurance quotes and TV adverts

How many months of lagged TV advertising should the model include?



poll.nectric.com.au/s/
africast-regression

0 responses

CLOSED 0 connected

Example: Insurance quotes and TV adverts

```
glance(fit_lags) |>
  select(.model, adj_r_squared, CV, AIC, AICc)
```

```
# A tibble: 4 × 5
  .model adj_r_squared    CV    AIC   AICc
  <chr>          <dbl> <dbl>  <dbl>  <dbl>
1 lag0           0.866 0.864 -3.38  -2.65
2 lag1           0.865 0.901 -2.17  -0.919
3 lag2           0.861 0.958 -0.208  1.73
4 lag3           0.859 0.995  1.35   4.15
```

Interpretation

No lag (`lag0`) has the lowest AICc and CV: quotes respond to advertising in the same month.

Example: Insurance quotes and TV adverts

```r
# Re-fit to all data
fit_adverts <- insurance |>
  model(TSLM(Quotes ~ TVadverts))
report(fit_adverts)
```

```
Series: Quotes
Model: TSLM

Residuals:
   Min     1Q Median     3Q    Max
-2.327 -0.543 -0.238  0.754  1.588

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   -0.240      0.901   -0.27     0.79
TVadverts      1.693      0.109   15.56   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.895 on 38 degrees of freedom
Multiple R-squared: 0.864,  Adjusted R-squared: 0.861
F-statistic:  242 on 1 and 38 DF, p-value: <2e-16
```

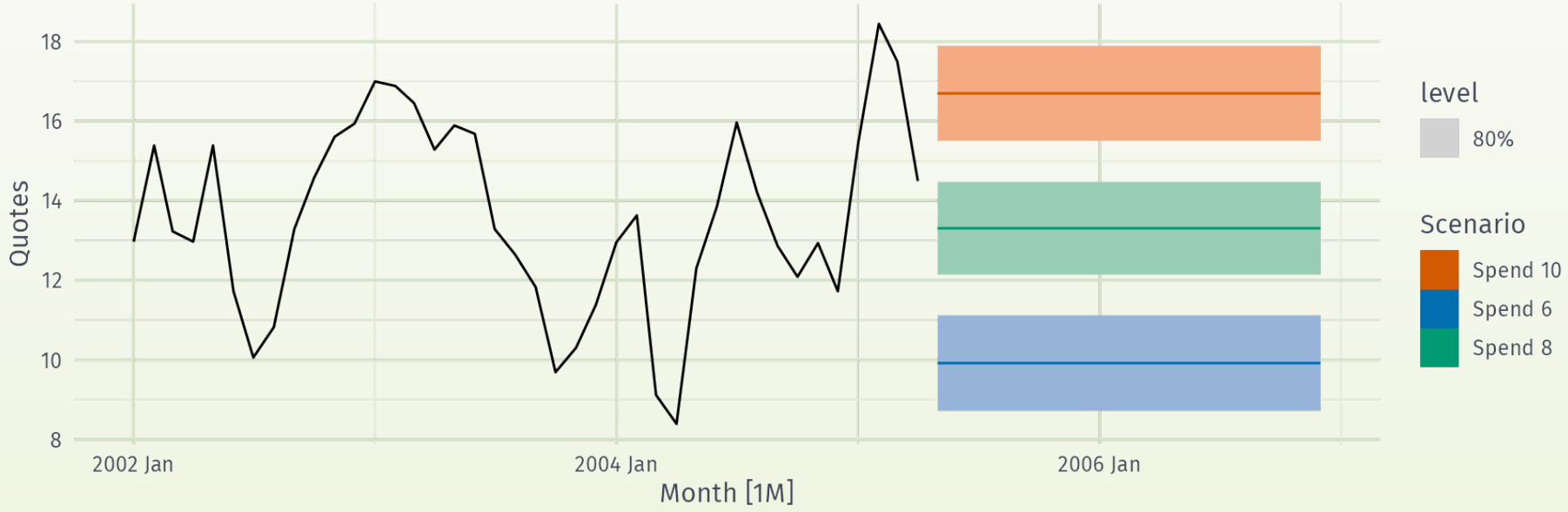
Interpretation

$\hat{y}_t = -0.24 + 1.69x_t$: each extra unit of spend brings about 1.7 more quotes that month.

Residuals are strongly autocorrelated (Ljung-Box $p < 0.001$), so prediction intervals will be too narrow.

Example: Insurance quotes and TV adverts

```
future_adverts <- scenarios(
  `Spend 10` = new_data(insurance, 20) |> mutate(TVadverts = 10),
  `Spend 8` = new_data(insurance, 20) |> mutate(TVadverts = 8),
  `Spend 6` = new_data(insurance, 20) |> mutate(TVadverts = 6),
  names_to = "Scenario"
)
insurance |> autoplot(Quotes) +
  autolayer(forecast(fit_adverts, future_adverts), level = 80)
```



Lead predictors

Sometimes y_t responds to a change in x before it happens, because the change is anticipated. Then y_t depends on future values $x_{t+1}, x_{t+2}, \ldots$

$$y_t = \beta_0 + \gamma_0 x_t + \gamma_1 x_{t+1} + \cdots + \gamma_k x_{t+k} + \varepsilon_t$$

- Create lead values by shifting the existing variable forwards with `lead()`
- e.g. sales y_t rise before an announced tax increase x_t takes effect
- Only usable for forecasting when the future x is known in advance (e.g. announced)